

Finding & Classifying Objects in Images

Computer Vision

Deep Learning

CNN

YOLO

Vision Transformers

What Is Object Detection?

Object detection is a computer vision task that simultaneously answers two questions about an image: *What objects are present?* (classification) and *Where are they?* (localization via bounding boxes). The output is a set of (class, confidence, x, y, w, h) tuples.

Relationship to RAG / LangChain / NLP

- **Core AI / Computer Vision** — Object detection is a CV task, not an NLP task. It is separate from RAG and LangChain.
- **Multimodal RAG** — In advanced pipelines, detected objects can be described as text and embedded into a vector store for visual Q&A.;
- **LangChain Vision Chains** — GPT-4o / Claude can be used inside a LangChain chain to describe images; object detection can pre-process frames before the LLM receives them.

Two-Stage vs One-Stage Detectors

Approach	How It Works	Examples	Speed / Accuracy
Two-Stage (Proposal-based)	Region proposals → classification of each region (ROI Pooling)	R-CNN, Fast R-CNN, Faster R-CNN	Slower, higher accuracy
One-Stage	Divide image into grid; predict boxes and classes in a single pass	YOLO (v3–v10), SSD, RetinaNet	Faster, real-time
Transformer-based	Detect objects as a set prediction problem with cross-attention	DETR, DINO, Grounding DINO	Flexible, strong accuracy

Key Concepts

- **Bounding Box** — Rectangle defined as (x_center, y_center, width, height) normalised to image size.
- **IoU (Intersection over Union)** — Measures overlap between predicted and ground-truth boxes. $\text{IoU} > 0.5$ typically counts as a correct detection.
- **NMS (Non-Maximum Suppression)** — Post-processing step that removes duplicate boxes by keeping the highest-confidence box and discarding overlapping ones.
- **Anchor Boxes** — Pre-defined box shapes the model predicts offsets from (used in YOLO, SSD).

- **mAP (mean Average Precision)** — Standard evaluation metric across classes and IoU thresholds.

YOLO — The Dominant One-Stage Detector

You Only Look Once (YOLO) divides the image into an $S \times S$ grid. Each cell predicts B bounding boxes with confidence scores and C class probabilities simultaneously — making it extremely fast (real-time on GPU). YOLOv8 / YOLOv10 are the current standard for production use.

Quick Start (YOLOv8):

```
from ultralytics import YOLO model = YOLO('yolov8n.pt') # nano = fastest results =  
model('image.jpg') # predict results[0].show() # visualise boxes
```

Vision AI Today

Foundation models like GPT-4o, Claude 3.5, and Gemini 1.5 can detect and describe objects in images without task-specific training, blurring the line between object detection and VLMs.